ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE F

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): design briefs and constraints, automating a bottle-filling station with a sequence and a ladder diagram, sensors and materials, project roles, budgets, drawings and procurement, testing and commissioning, and maintenance schedules, manuals and fault-finding.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw sketches, tables and circuits neatly. Use only a black or blue pen.

> **PROJECT.** A small juice company fills bottles by hand from a tank. The owner wants a low-cost automated filling station: bottles move on a conveyor, a sensor detects each bottle under the filling nozzle, the conveyor stops, a valve fills the bottle for a set time, and the conveyor starts again to move the full bottle away.

QUESTION 1: THE DESIGN

1.1 Explain the difference between a design brief and a design specification. (4)

1.2 Name THREE constraints that could limit the design of the filling station. (3)

1.3 Explain why the people who will use and maintain the machine must be considered during the design. (3)

[10]

QUESTION 2: AUTOMATING THE FILLING STATION

2.1 Make a neat freehand sketch of the automated filling station. Show the conveyor, the bottle sensor, the filling nozzle and valve, the juice tank and the drive motor. (5)

2.2 Write the sequence of operations of the filling station as numbered steps, from START to the full bottle moving away. (6)

2.3 Draw a ladder diagram for the filling station. The conveyor motor (M) runs while the station is switched on (START, STOP and latching); when sensor S1 detects a bottle, the conveyor stops and the filling valve Y1 opens for 4 seconds (timer T1), after which the conveyor runs again. (8)

2.4 Name FOUR types of sensors that could detect a bottle on the conveyor. (4)

2.5 Explain why stainless steel is used for the parts that touch the juice. (2)

[25]

QUESTION 3: PROJECT MANAGEMENT

3.1 Explain the role of EACH of the following members of the project team: project manager; designer; technician; quality controller. (8)

3.2 Draw up a budget template for the project with suitable column headings and at least FIVE items. (8)

3.3 Explain what a part drawing and an exploded (assembly) drawing are. (4)

3.4 List FIVE safety precautions to follow while assembling the filling station. (5)

3.5 Explain the terms procurement and lead time, and why lead time matters in project planning. (5)

[30]

QUESTION 4: TESTING AND COMMISSIONING

4.1 Explain the difference between functional testing and performance testing. (4)

4.2 List THREE faults that would cause a test run to be stopped. (3)

4.3 Explain the purpose of simulating the PLC program before running the machine. (2)

4.4 Explain how the filling time would be adjusted during commissioning so that each bottle contains the correct volume. (3)

4.5 Explain why the client signs a commissioning report when the machine is handed over. (3)

[15]

QUESTION 5: MAINTENANCE

5.1 Draw up a preventive maintenance schedule for the filling station with FOUR items, the task for each and how often it must be done. (8)

5.2 Explain how to test a PLC input sensor during maintenance. (3)

5.3 List FOUR sections that should appear in the index of the technical manual for the filling station. (4)

5.4 Explain what an open-circuit fault is, and give ONE way to find it. (3)

5.5 Give ONE advantage of monitoring the filling station remotely. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER